

Trainer Debrief — Lab 30

Financial Benefits Validation, Handover and Project Closure

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Validate benefits with Finance and formally close your capstone project (A5, K1).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-30-benefit-validation.xlsx

Hard benefit = scrap 172k + rework 89k + warranty 116k + freight 56k + overtime 54k = SGD 487,000 per year. The SGD 54k of 'lost margin' is SOFT (cost avoidance / capacity) and must be reported SEPARATELY — claiming it as hard savings is the single fastest way to lose Finance's sign-off. Implementation cost = SGD 168,700. Simple payback = 168,700 / 487,000 ~ 0.35 years (about 4.2 months). Year-1 net benefit ~ SGD 318,300. Every line must name its validation SOURCE and be signed off by the Finance controller — an unsigned benefit is an estimate, not a benefit.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Your benefit is separated into hard, soft and cost-avoidance components, validated statistically and signed off by Finance, and the process owner has explicitly accepted handover with a scheduled sustainability audit.

Where this sits in the capstone

Benefits are not real until Finance signs them. This lab converts your improvement into a validated financial number, transfers ownership to the process owner, and formally closes the project — the step most often skipped, and the reason gains quietly evaporate. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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